

Track1_nottingDuck: Our samantha – a useful shell tool

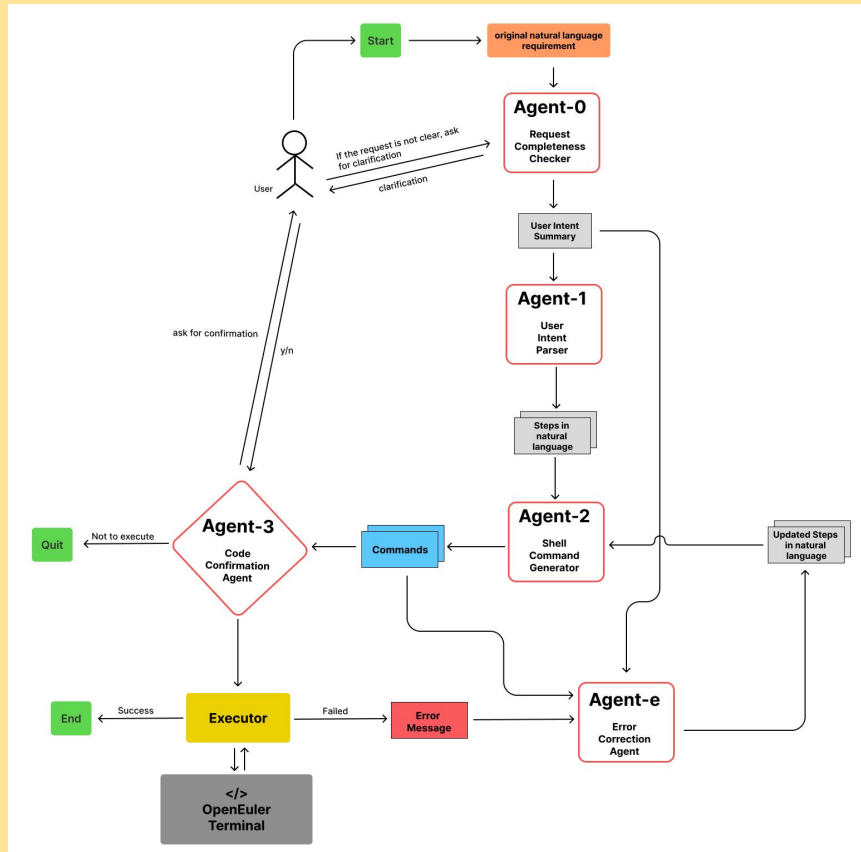
Overall structure

- input: user's natural language
- totally 5 agents
 - agent_0: ensure clear user input
 - agent_1: break input into action steps
 - agent_2: command code generator
 - agent_3: ask user for confirm
 - agent_e: error handling

current result:

- Achieve all the functions in Tier1 and Tier2.
- Achieve most of functions in Tier3

Tier	Category	Tasks	Check
Tier 1	Navigation	Change directory	
		List files	
	Creation	Create new empty file	
		Create new empty directories	
	Basic Manipulation	Copy file/directory	
		Move file/directory	
		Rename file/directory	
	Simple Search	Find file/directory by their exact name	
	User Feedback	Provide clear confirmation of actions taken or clear error messages	
	Critical Feature Safety	must implement a confirmation step for any destructive operations like deleting files or directories	
Tier 2	Advanced Search & Filtering	Find files by type	
		Find files by extension	
		Find files by size	
		Find files by modification date	
	Contextual Understanding	Understand relative paths implicitly	
	Basic Multi-step Operations	Handle a simple sequence of actions in a single command	
Tier 3	Robust Error Handling	Gracefully handle incorrect paths, permission issues, or non-existent files within the openEuler file system.	
	Content-Aware Search	Search for files containing specific keywords	
	Agentic Capabilities	Attempt to resolve different types of real-life problems using a combination of tools	
	Self-Correction	The agent workflow does not exit when an error is encountered, instead this OS Assistant will try different approaches to achieve the goal.	
	Organizational Intelligence (Bonus)	Proactively suggest organizational actions	
	Natural Conversation Flow (Bonus)	Handle follow-up questions and clarifications	



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Usages/result:

Following are how we currently match the challenge Tier

- basic file operations:
 -  **Navigation:** change directories and list files using natural language.
 -  **Creation:** create new files and directories.
 -  **Basic manipulation:** copy, move, and rename files or directories.
 -  **Simple search:** find files or directories by exact name.
 -  **User feedback:** provide clear confirmation of actions or error messages.
 -  **Safety mechanism**
- enhanced intelligence:
 -  **Advanced search and filtering** by type, extension, size, or modification date.
 -  **Context awareness:** understands relative paths and context implicitly.
 -  **Basic multi-step operations:** handles sequences of actions in a single request.
 -  **Robust error handling:** gracefully handles invalid paths, permission issues, and missing files.
- more advanced:
 -  **Content-Aware Search:** able to search for files containing specific keywords.
 -  **Self-Correction:** when an error is encountered, Samantha will try different approaches to achieve the goal. (Up to *three* self-correction attempts will be tried)
 -  **Natural Conversation Flow:** Samantha can handle follow-up questions and clarifications. In case the request is not clear, she will ask for clarification until she exactly gets your point. If during the conversation you don't want her to execute the request anymore, she will simply do nothing and exit.

```
[root@a2e09f9b33eb work]# samantha delete test.txt in test folder
Checking the clarity of your request...
Agent: After completion, should we stay in the starting directory
or stay in the destination directory?
User: stay
Trying to understand your request...
Generating commands...
# Command Explanation
1. Go to the "test" folder inside the current directory.
2. Delete the file named "test.txt" inside that folder (if it exists).
3. Return to the previous directory (back to /work).

# Danger Warnings
- ⚠️ The file "test.txt" will be permanently deleted without confirmation if it exists.

# Confirmation
Do you want to proceed? (y/n)
y
Commands executed successfully ^-^
-----Output-----
```

a simple example (red arrow shows the user input)

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Future work

Aims to achieve full functions in Tier3

- **A logging system (already implement):** record user interactions and execution information. This provides the groundwork for **context-aware decision-making**.
- **A vector space database:** includes a reserved vector space database interface, enabling future integration of **semantic search** for PDF files and images.
- **A Tool Integration Pathways:** an inclusion of external tools, allowing the assistant to **extend its capabilities** beyond basic shell commands. This system will be achieved by using LangGraph.

Possible powerful future application:

- samantha set up and configure cuda environment in my computer
- samantha automatically tidy up my desktop
- ...

